

Erdős Problem #394

EP-394

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $t_k(n)$ denote the least m such that $n \mid m(m+1)(m+2)\cdots(m+k-1)$. Is it true that $\sum_{n \leq x} t_2(n) \ll \frac{x^2}{(\log x)^c}$ for some $c > 0$?

Is it true that, for $k \geq 2$, $\sum_{n \leq x} t_{k+1}(n) = o\left(\sum_{n \leq x} t_k(n)\right)$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>